

Attestor

A cross-chain loan repayment history you cannot lie to — built on Attestcoin.

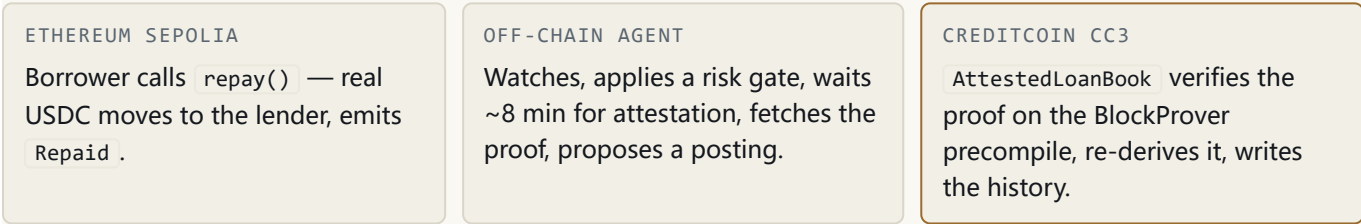
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THE PROBLEM

Creditcoin exists to make credit history verifiable. But the money that builds that history — stablecoin loans and repayments — lives on Ethereum. Bridging the *record* across chains usually means trusting a relayer's word that a repayment happened. A credit history is worthless if it can be forged.

THE SOLUTION

A borrower repays on Ethereum; an autonomous agent posts it to a credit history on Creditcoin — but the record is written **only** if an Attestcoin proof shows the repayment really happened. The agent proposes; the on-chain book disposes. **Neither the agent nor we can write a false entry.**



Remove Attestcoin and there is no product: the book holds no trusted input of its own. Every write to a credit history is downstream of one precompile call — `0x...0FD2 . verify(...)` returning without reverting.

THE SECURITY MODEL — ATTESTCOIN PROVES LESS THAN IT APPEARS

It proves a transaction was **included in a confirmed block** — not that it *succeeded*, called the right contract, or was sent by who it claims. The book enforces, on-chain, every property the proof does not:

CHECK	WITHOUT IT
<code>status == 1</code>	a reverted repayment posts as if it paid
<code>chainId (resolved)</code>	a valid proof from the wrong chain is accepted
<code>source contract</code>	any contract's call is posted as a repayment
<code>repay selector</code>	a different function passes as a repayment
<code>borrower == sender</code>	credit is assigned to whoever relays the proof
<code>freshness + maxAge</code>	an attested repayment is postable forever
<code>tx-hash consumed</code>	one repayment is posted again and again

Plus economic guards: zero-amount and self-payment repayments are refused on both chains, because an entry is only worth something if real money moved. **28 contract tests; 22 reject a forgery.**

WHAT WE FOUND THAT THE DOCS DON'T SAY – SEVEN UNDOCUMENTED PROTOCOL FACTS

- 1 **The precompile reverts on a bad proof; it does not return `false`** as the SDK documents — so `if (!verified)` is dead code.
- 2 **~8-minute latency is a real constraint:** a 32-block reorg window named only in an error string, plus ~39 blocks of attestation lag. We surface it, not hide it.
- 3 **`chainKey` is environment-scoped** — key 1 is Sepolia on testnet, Ethereum mainnet on mainnet. Bind to `chainId`, or a promotion silently changes trust.
- 4 **SDK and chain names differ:** the function is `verify` on-chain, not the SDK's `verifySingle`; `ChainInfo` is `snake_case` on-chain.
- 5 **The proof envelope is `(uint8,bytes[])`, not `bytes[]`** — and the SDK's own docs say the wrong one. A consumer written from the docs decodes nothing.
- 6 **The precompiles can't be fork-tested, and the failure is silent** — a call to the codeless address succeeds and returns nothing, passing the wrong assertion.
- 7 **`get_chain_by_key` returns one tuple, not two values** — a bug we shipped; every mocked test passed, it reverted every time on-chain. It's why we built a live-check layer.

Findings 5 and 7 make a documentation-following consumer fail outright. They are our ecosystem contribution as much as the product is.

LIVE & VERIFIABLE – NO KEY, NO GAS

reproduce	<code>node spike.mjs</code>	MockUSD	<code>0xCFd5E8e6...d0ef</code> (Sepolia)
tests	<code>forge test</code> – 28 pass	LoanBook	<code>0xe31906a2...5e9</code> (CC3) · LoanRepay
live	<code>node tools/live-check.mjs</code>	<code>0x08F8b91A...</code> (Sep)	
agent	20 tests, tsc clean	repay (Sep) <code>0x49592b0c...</code>	post (CC3) <code>0x0f3d4ca0...</code>
		BlockProver <code>0x...0FD2</code>	ChainInfo <code>0x...0FD3</code>

HONEST STATUS & ROADMAP

The full loop is live, on-chain both sides. A borrower repaid 250 mUSD on Sepolia (`0x49592b0c...`); the agent proved it and posted it to the loan book on Creditcoin (`0x0f3d4ca0...`), which now reads `totalRepaid 250000000`, `repaymentCount 1` for the borrower. Nothing is mocked — reproduce with `node tools/post-once.mjs`.

Next: the agent running continuously as a service · multi-loan credit scoring read by a Creditcoin lending contract · batch proofs (the SDK supports up to 10) · mainnet with real USDC and a governance-set source binding.